Venue, Resource and Staff Allocation System

Event Management Domain

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Table of Contents:

1 INTRODUCTION.....1

1.1 PURPOSE...…….....1

1.2 SCOPE....……..1

1.3 NEED FOR THE SYSTEM….....1

1.4 GLOSSARY……2

1.5 REFERENCES…..……..2

2 USER REQUIREMENT……3

3 SYSTEM REQUIREMENT……4

3.1 FUNCTIONAL REQUIREMENT……..4

3.2 NONFUNCTIONAL REQUIREMENT……..5

4 **SYSTEM MODELS ……...6**

4.1 USE-CASE DIAGRAM……6

4.2 ACTIVITY DIAGRAM……..7

4.3 SEQUENCE DIAGRAM……13

INTRODUCTION

Purpose

The purpose of the Resource, Staff, and Venue Allocation System is to support efficient planning, coordination, and management of events by organizing the allocation of venues, resources, and staff. The system aims to reduce scheduling conflicts, minimize manual errors, improve communication among stakeholders, and ensure smooth execution of events. It also helps in monitoring availability, handling last-minute changes, and supporting contingency planning to enhance overall event reliability

Scope

The Resource, Staff, and Venue Allocation System is designed to support the planning, scheduling, and management of events by organizing the allocation of venues, resources, and staff. The system allows clients to create event requests, vendors to manage availability and allocations, staff to view assignments, and administrators to monitor and supervise operations.The system provides features such as availability checking, conflict detection, resource and staff assignment, scheduling support, reporting, and basic communication. It also supports buffer and contingency planning to handle unexpected changes and operational risks.

The scope of the system is limited to managing pre-event planning, event-day coordination, and post-event updates. It does not include physical execution of event activities, financial accounting systems, or third-party logistics management

Need for the System

The need for this system arises from the challenges of manually managing staff, resources, and venues, which leads to conflicts, delays, and inefficiencies.There is a need for a centralized system that can organize event information, monitor availability, detect conflicts, and support effective coordination among clients, vendors, staff, and administrators. The proposed system aims to reduce manual effort, improve accuracy, enhance transparency, and ensure smooth execution of events.

Glossary

Admin	User who manages the entire system
Vendor	Provides venues and event resources
Client	User who request and organizes events
Staff	Personnel assigned to events
Resources	Equipments used in events(chair, lights, sound system)
Allocation	Assignment of resources or staff events
Event	A planned function such as wedding or conference
Buffer Time	Extra time kept for setup, delays, or emergencies.
Contingency Plan	A backup plan used when unexpected problems occur.

References

1.Interview and requirements discussion with **Ms. Gomathi Ramesh**, Event Management Professional, *Bluesky Learning / Success Gyan India LLP*,
-Interaction on Feb 03 2026. Documented in *interactions.md*.

2.Online interaction and requirement document shared by **Mr. Raj Hansoge**, Managing Partner, *EDICO / Buenos Productions*,
- Interaction on Feb 01 2026. Compiled in *interactions.md*.

User Requirements

Admin

1. Manage users, venues and resources.
2. Preset Creation.
3. View reports and system data.

Vendor

1. Approve or reject the event request
2. Manage available resources.
3. Assign staff and resources to events.
4. Vendor-to-vendor communication (for coordination)
5. Vendor-client communication
6. Automated overlapping and conflict detection
7. Post-event resource release and update

Client

1. Create event requests
2. Select the venue and required resources.
3. View event status.

Staff

1. View assigned tasks and shall accept or reject it.
2. Update availability
3. Operational Status & Progress Updates
4. View work history.

SYSTEM REQUIREMENT

Functional Requirements

User Authentication

1. Users can register and log in based on their assigned roles (Admin, Vendor, Client, Staff).
2. Authenticate users before granting access to system features.

Event Management

1. Clients create event requests by selecting venue, date, and required resources.
2. Vendors can review and approve or reject event requests.
3. Clients can view the status of their event requests.

Venue and Resources Allocation

1. Admin to add, update, and manage venues and resources.
2. Allocate venues and resources to approved events.
3. The system shall prevent double booking of venues and resources.

Staff Management

1. Vendors can assign tasks to staff based on availability.
2. Staff members can update their work progress.
3. Maintain staff schedules and task status.

Non-Functional Requirements

Performance

1. Process event request and allocation with minimal delay
2. Support multiple concurrent users

Security

1. Protect user data using secure authentication
2. Ensures role based access control

Usability

1. Have an user friendly interface
2. Require minimal training fo risers

Scalability

1. Support an increasing number of users, event and resources
2. Allow future enhancement without major redesign

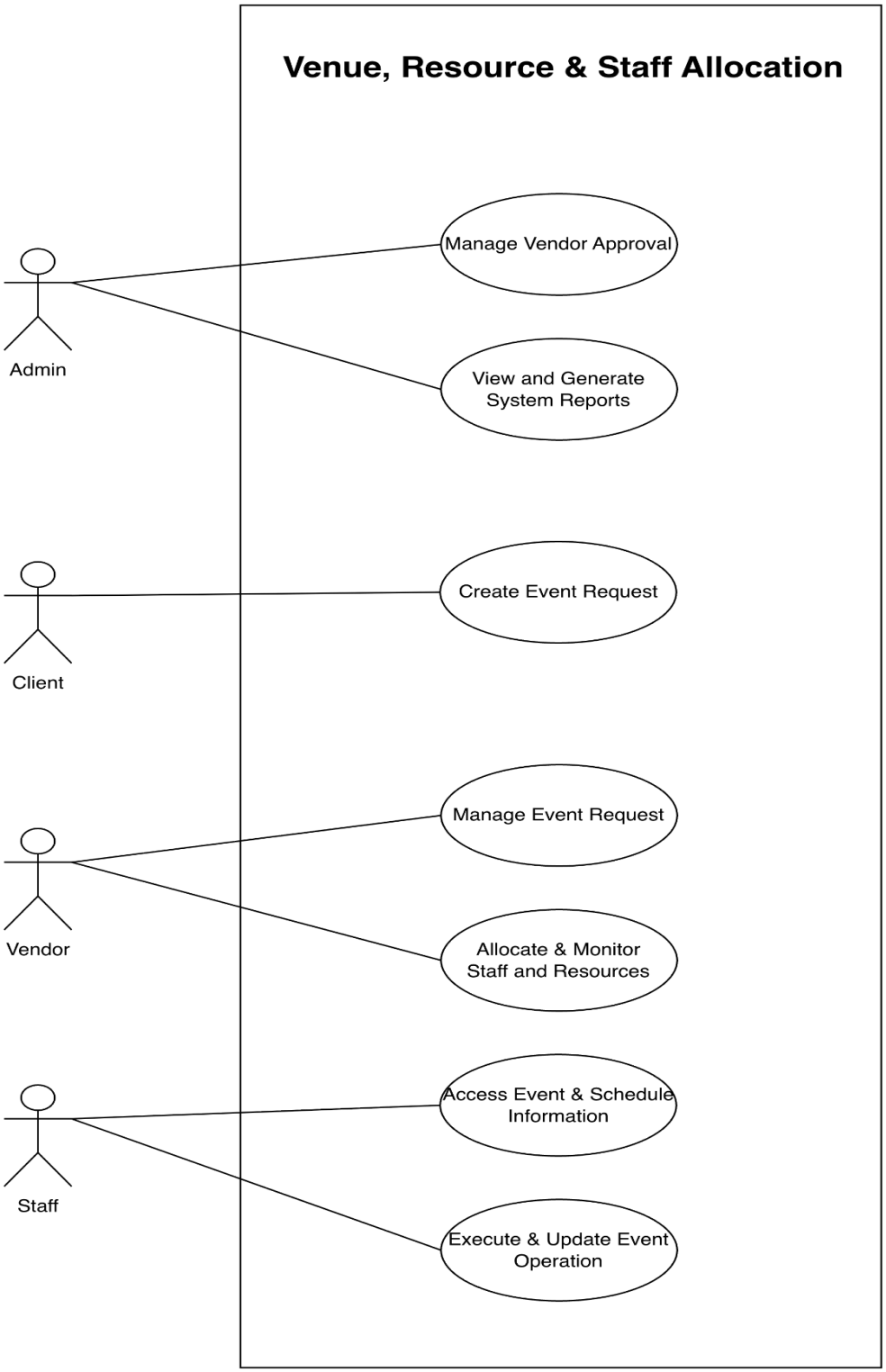
Maintainability

1. Easy to update and maintain
2. Support modular upgrades and bug fixes

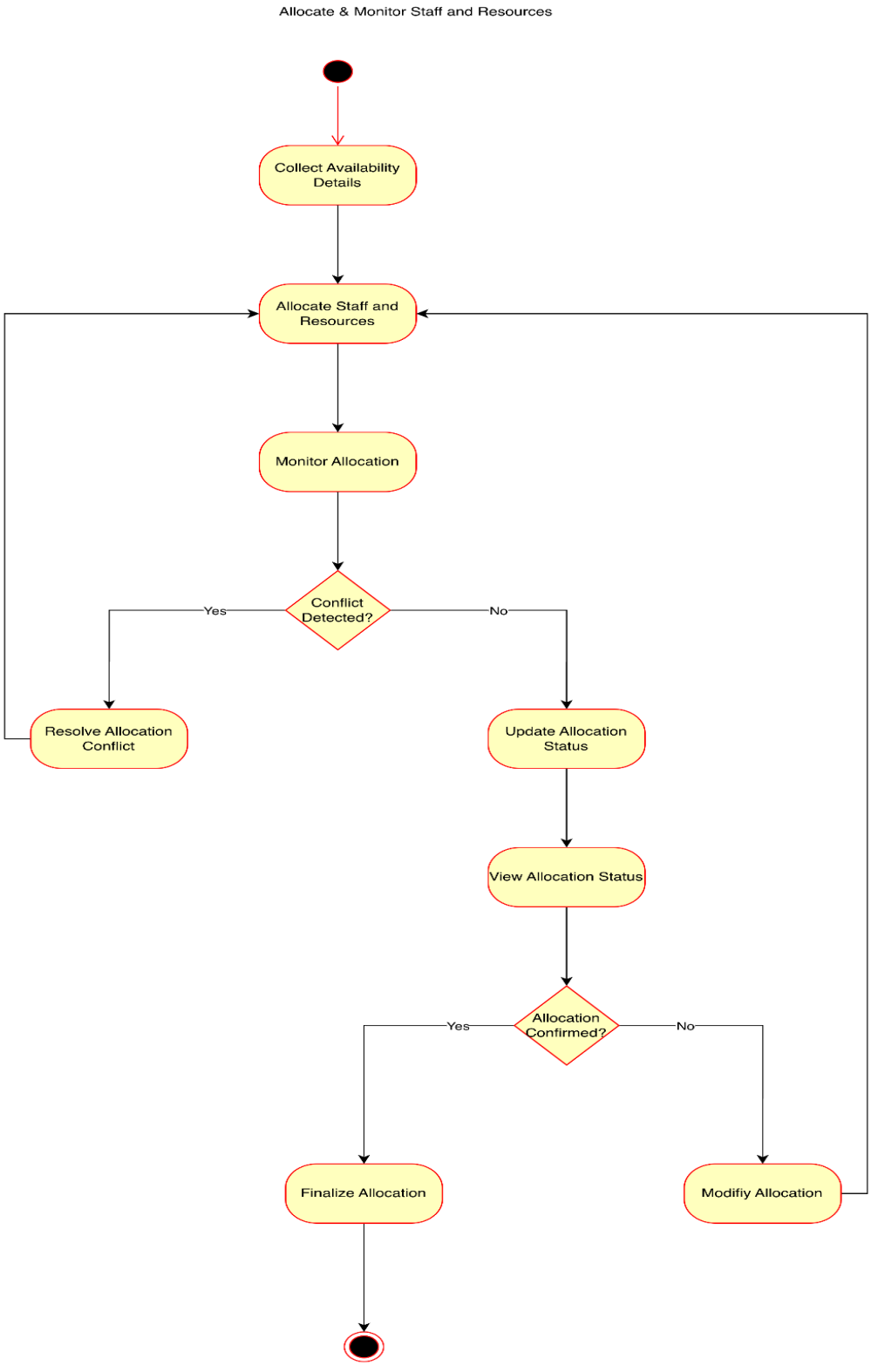
Dependability

1. Maintain high availability of system during normal operation
2. Prevent data loss in case of system failure

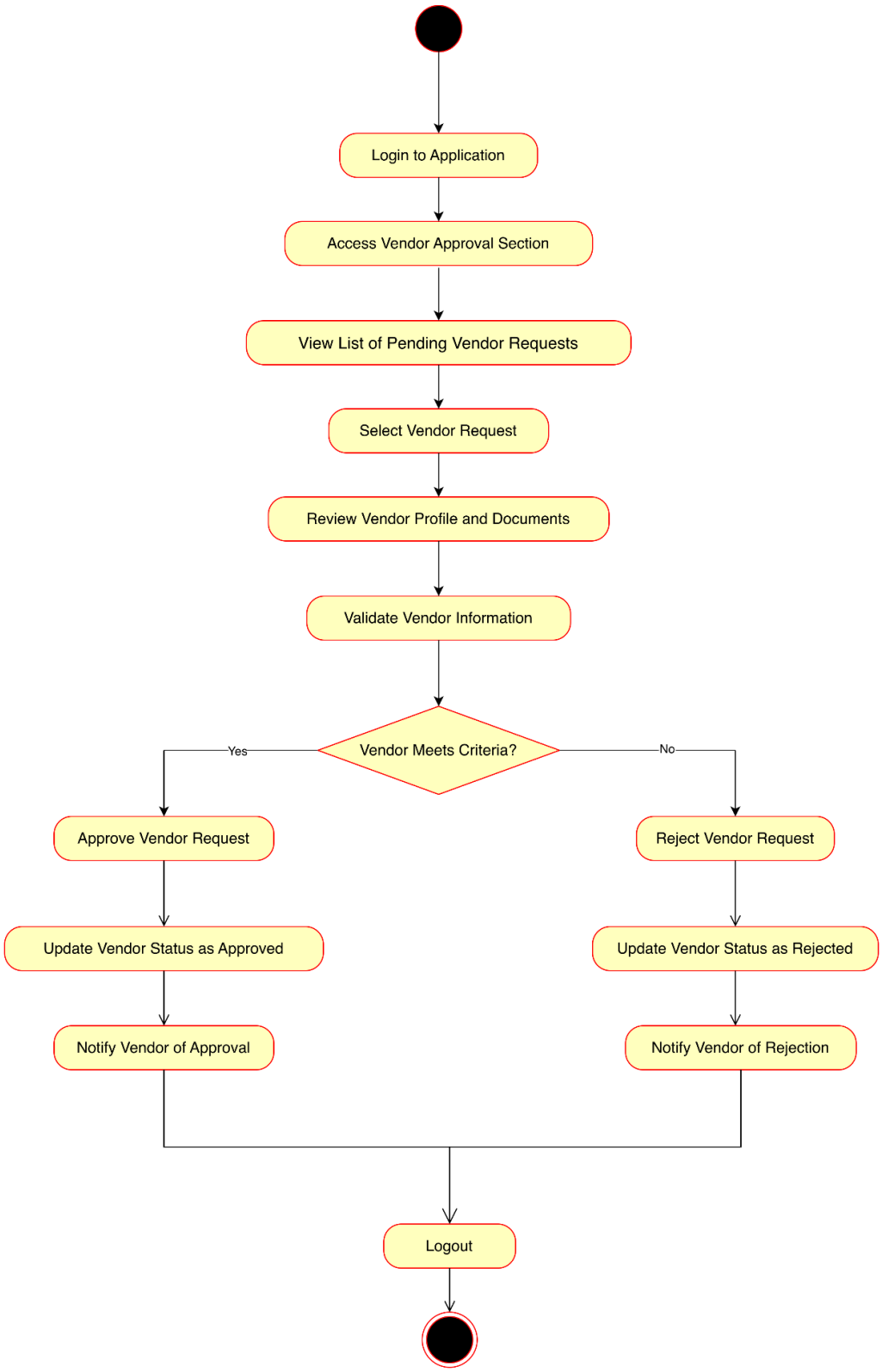
Use Case Diagram



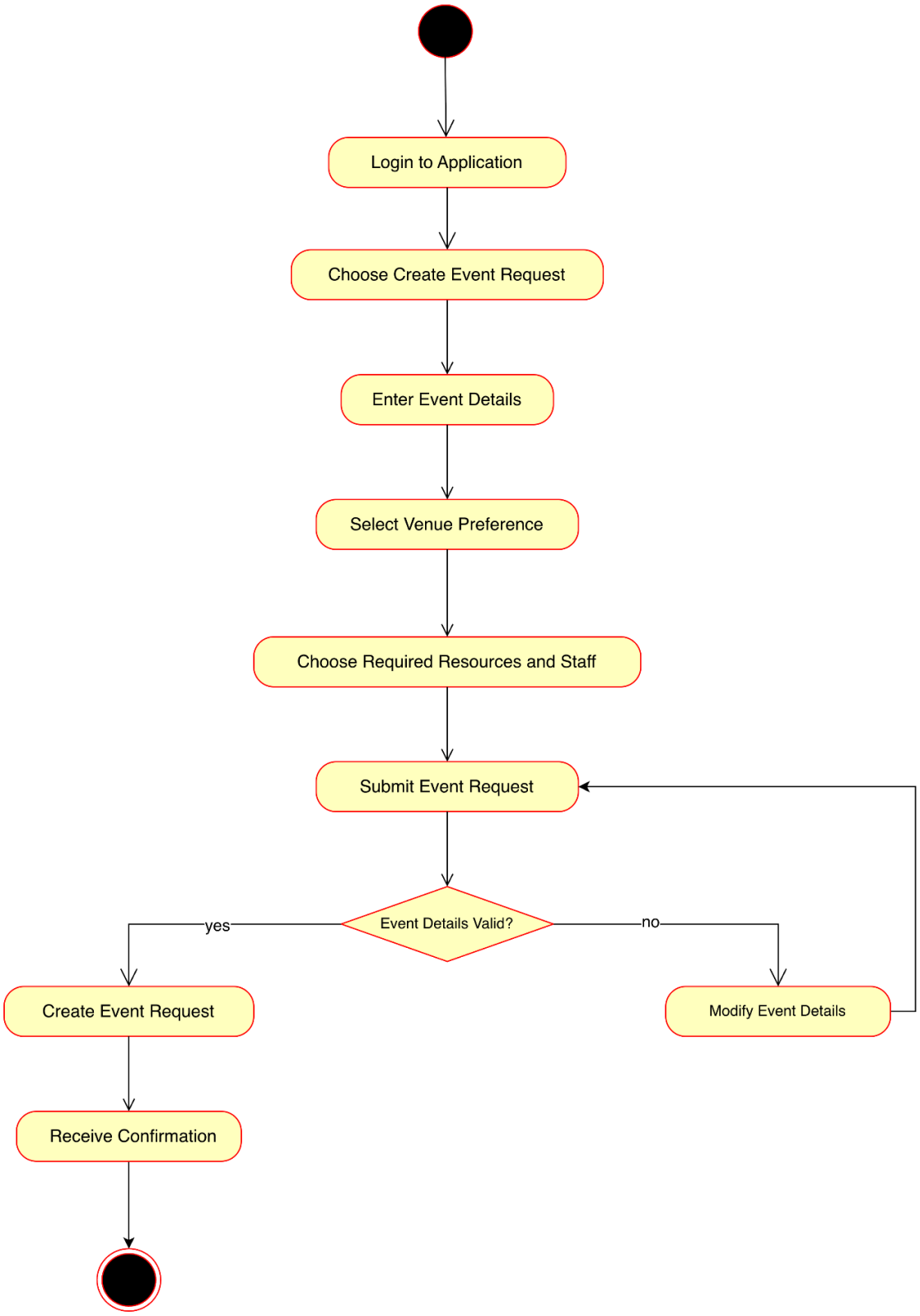
Activity Diagrams



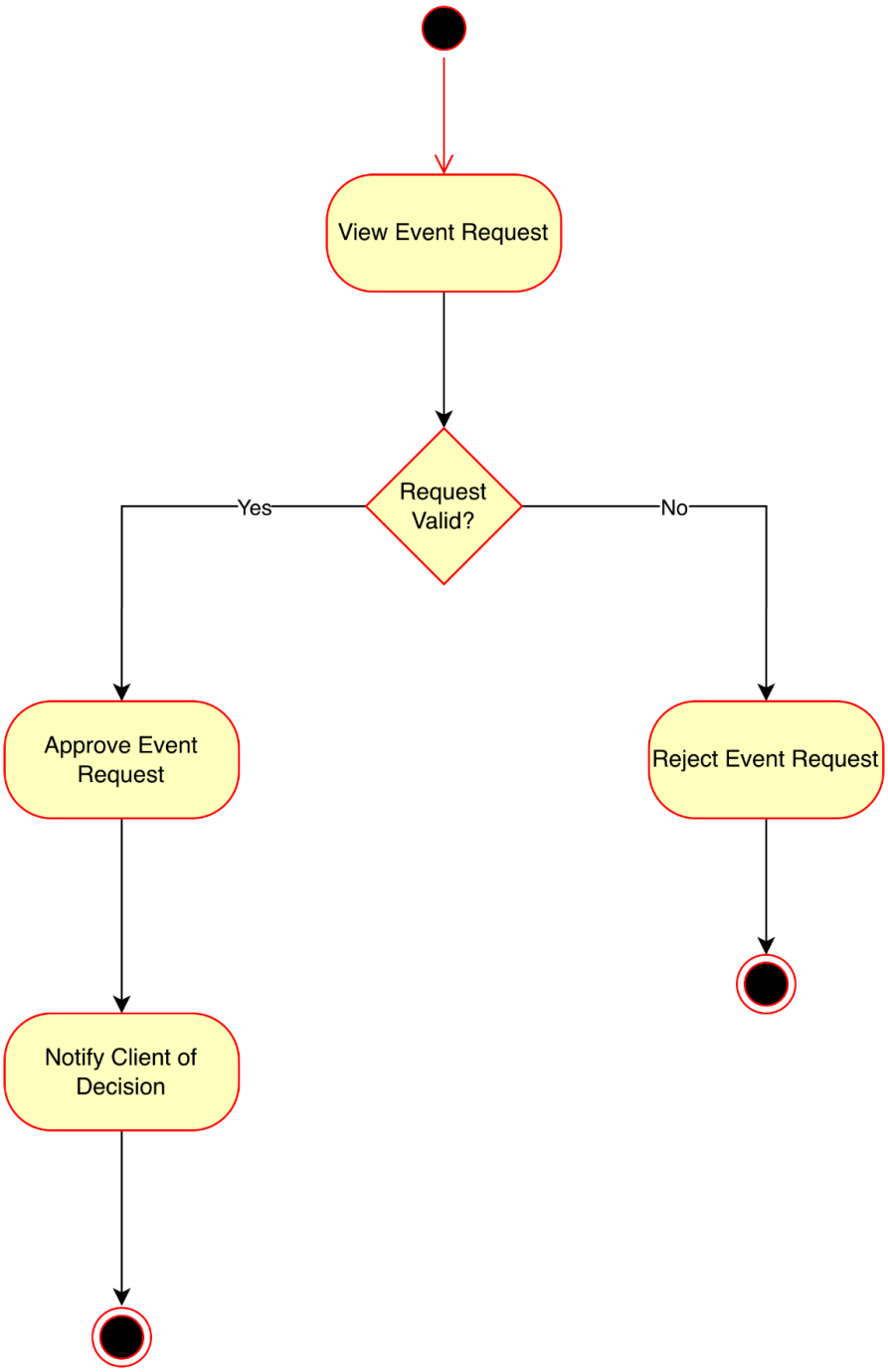
Manage Vendor Approval



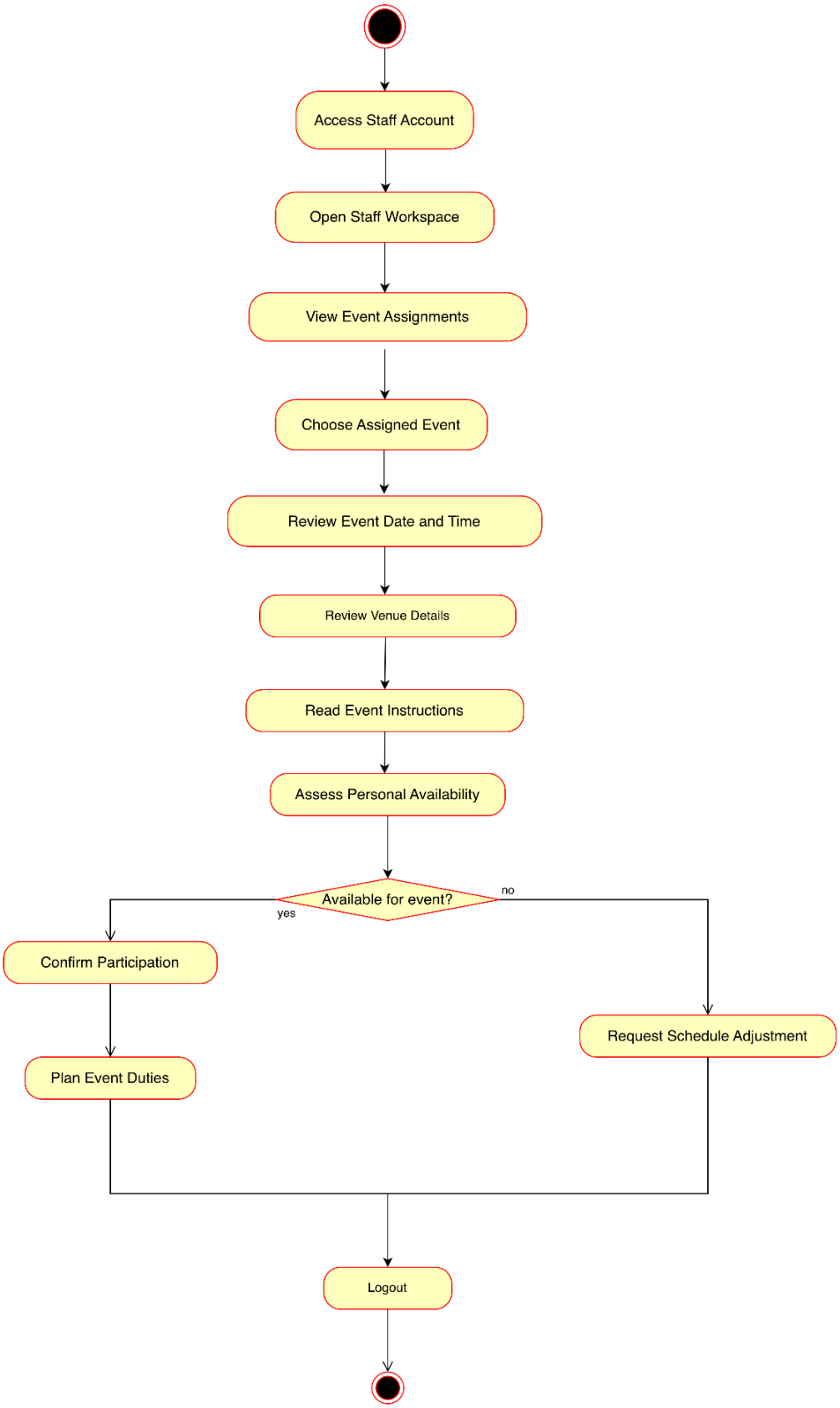
Create Event Request



Manage Event Request

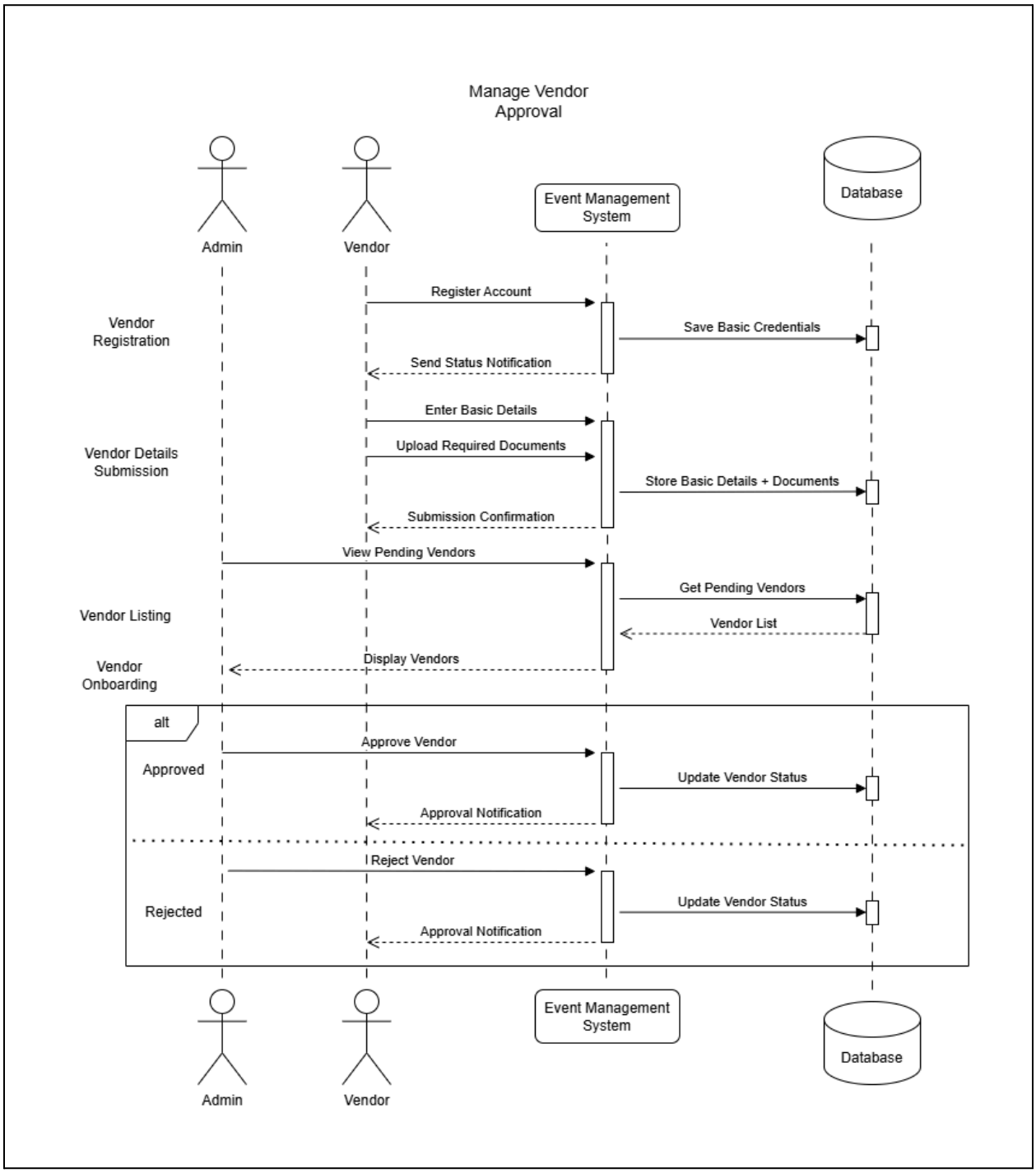


Access Event & Schedule Information

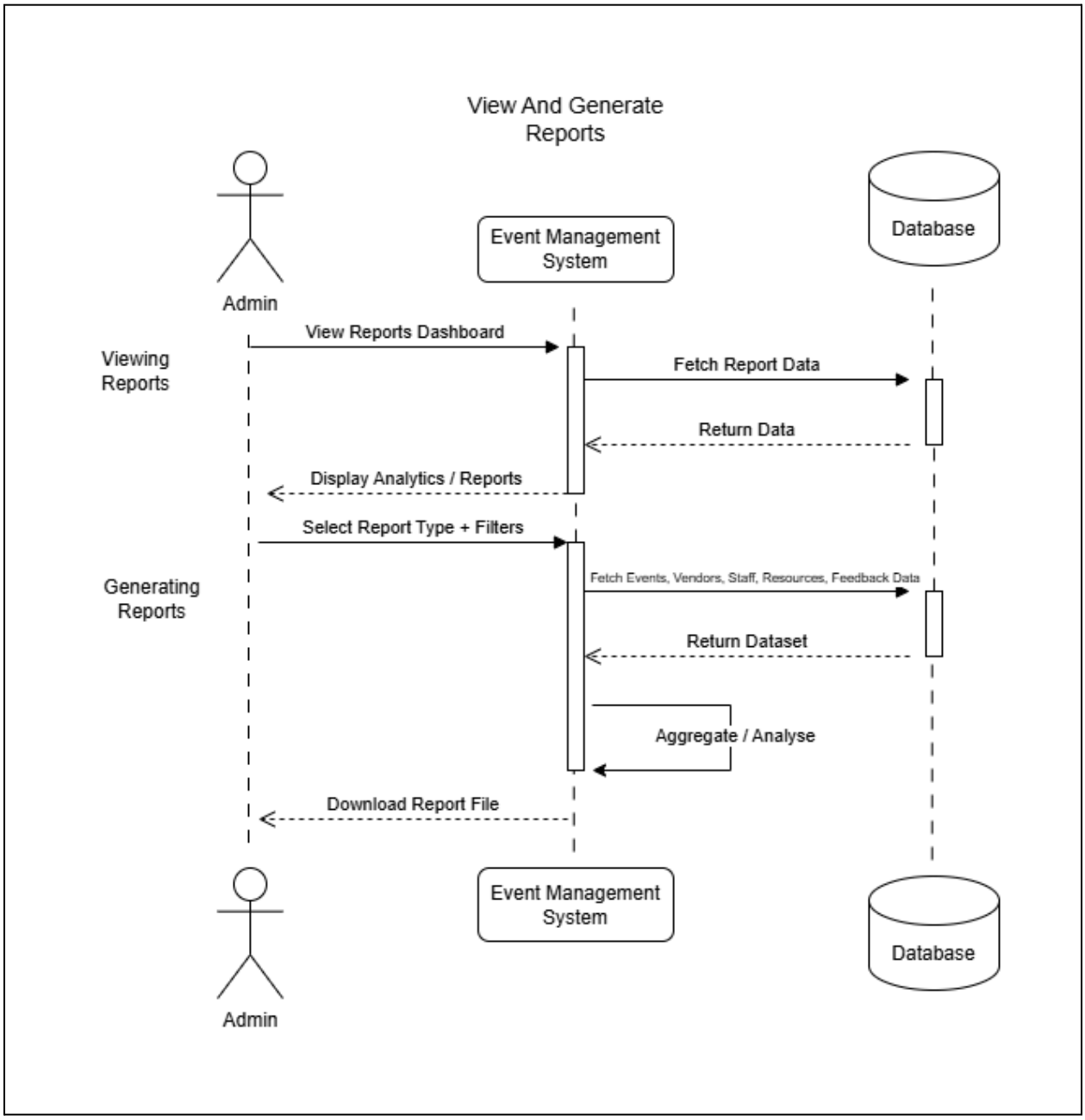


Sequence Diagrams

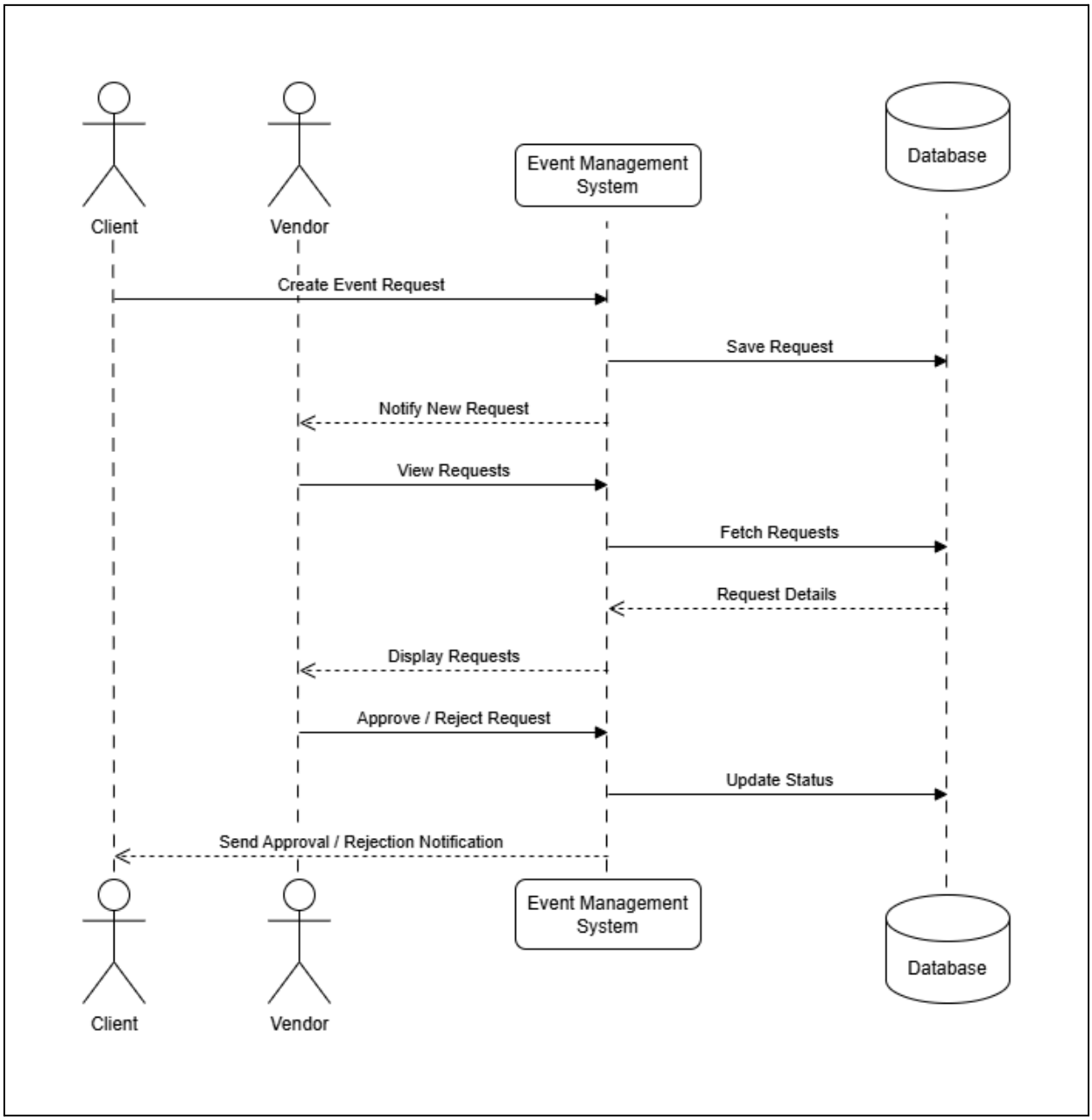
1. Manage Vendor Approval

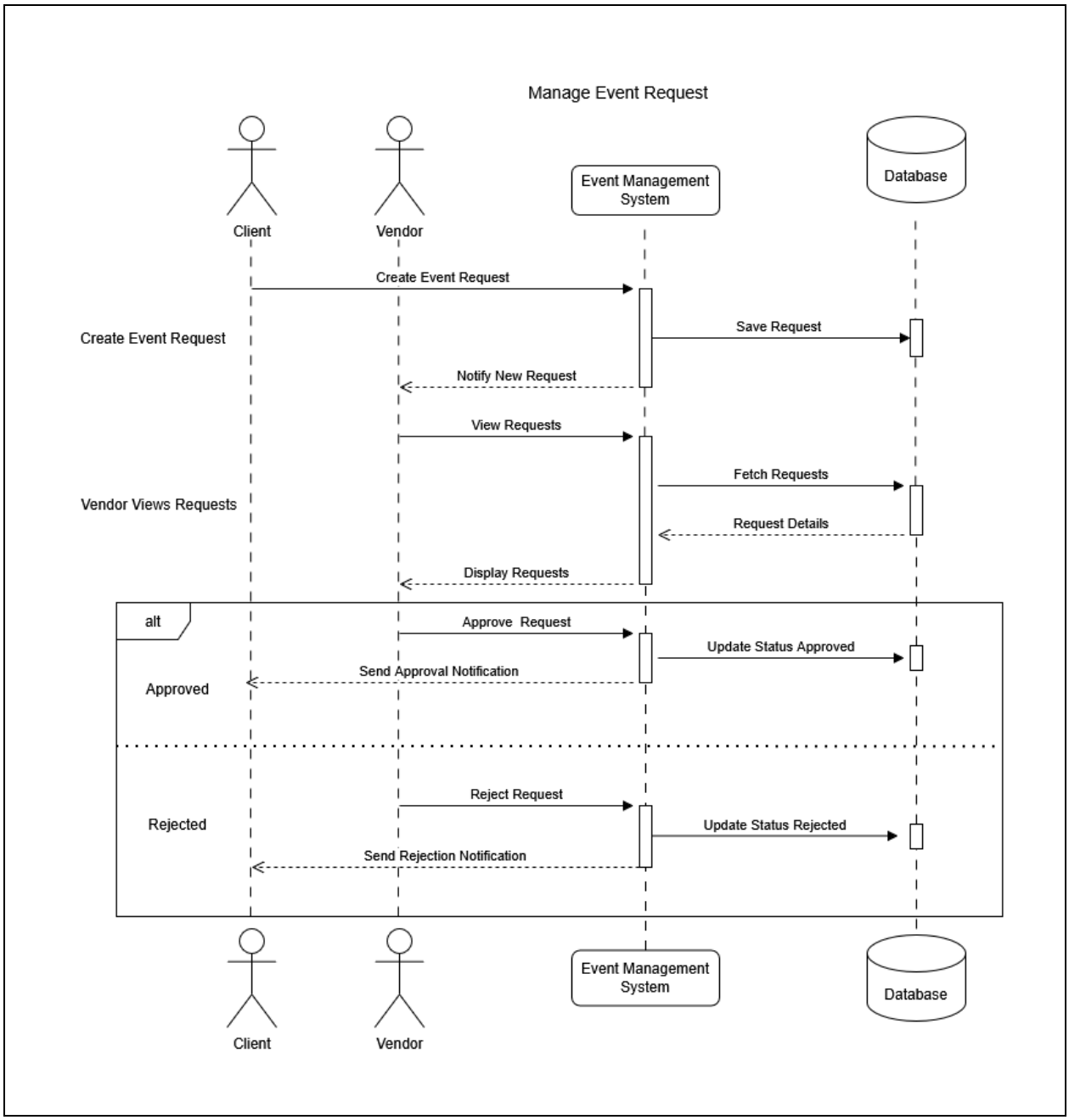


2. View and Generate Reports:

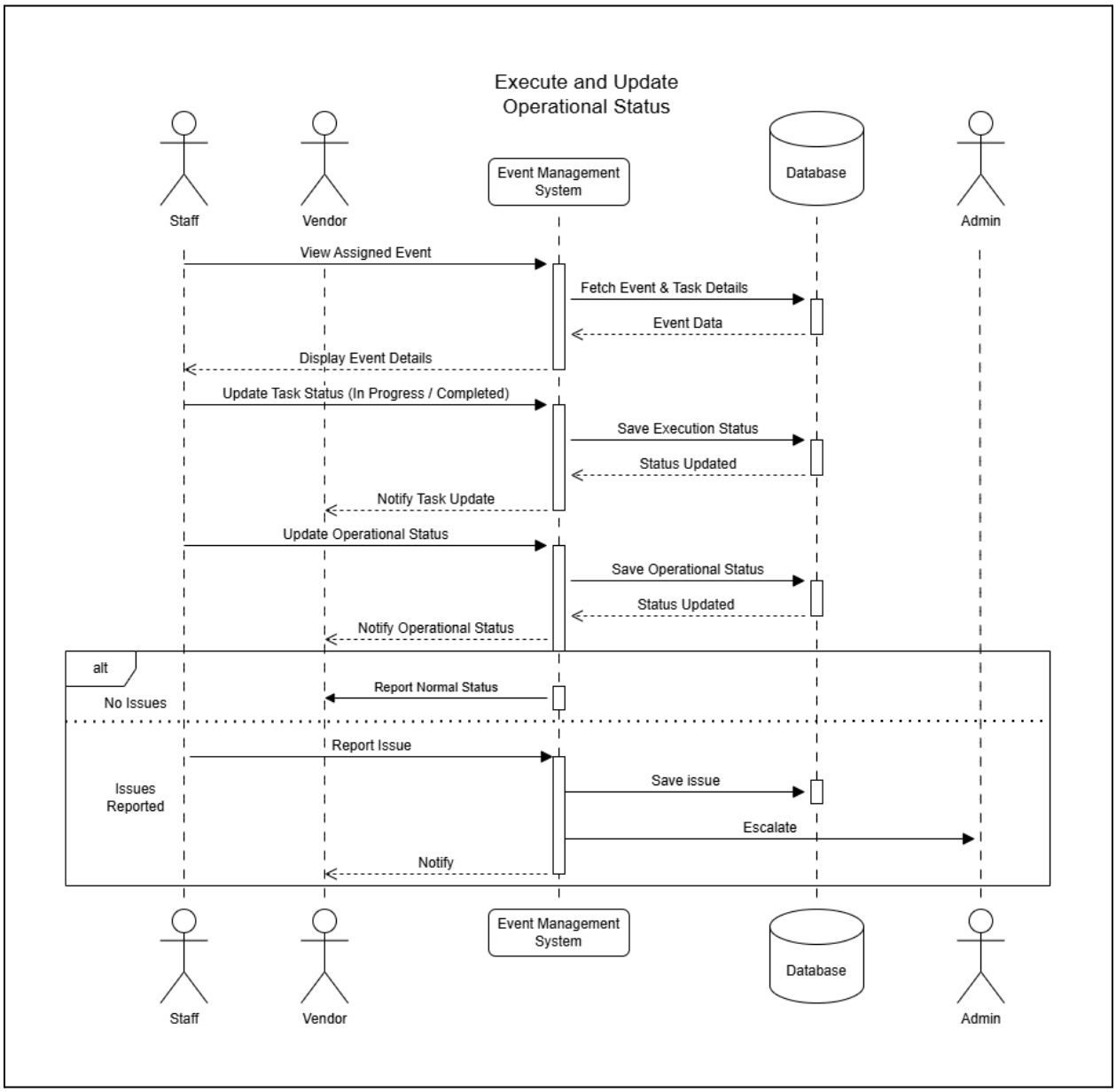


3. Manage Event Requests:




Manage Event Request
Client
Vendor
Event Management System
Database
Create Event Request
Create Event Request
Save Request
Notify New Request
View Requests
Fetch Requests
Vendor Views Requests
Request Details
Display Requests
alt
Approve Request
Update Status Approved
Send Approval Notification
Approved
Reject Request
Rejected
Update Status Rejected
Send Rejection Notification
Client
Vendor
Event Management System
Database

4. Execute and Update Operational Status:



5. Staff and Resource Allocation:

